

The Conative-Integrative Framework (CIF)

How Active Inference Networks, Integrated Information (Φ), and the 6th Axiom Fit Together to Unite Analytic Idealism, the Free Energy Principle, and Consciousness

Author: Thomas Riebl (Luxembourg)

Date: August 30, 2026

Theoretical Architecture: The Conative-Integrative Framework (Analytic Idealism $\times$ Free Energy Principle $\times$ Expanded IIT 4.0 $\times$ Ego Tunnel Theory $\times$ Tri-Level Brain Architecture)

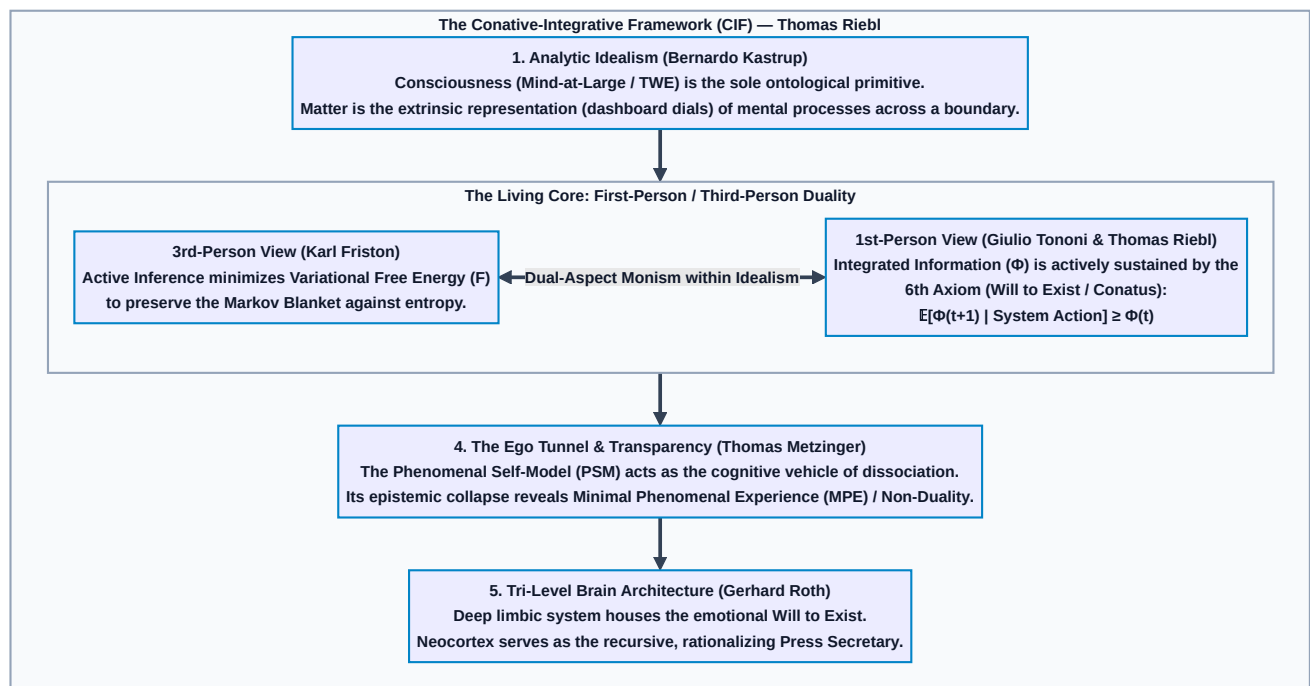


Table of Contents

- Executive Summary & Theoretical Thesis
- 1. The 1st-Person / 3rd-Person Equivalence: Friston Meets Tononi & Riebl
- 2. The Mechanism of Dissociation in Analytic Idealism
- 3. Demolishing the Paradox of “Transient Causal Phantoms”
- 4. The Epistemology of Self-Inquiry: Metzinger, Roth, and the Mirror of Mind
- 5. Computational and Experimental Validation
- 6. Ontogenetic Scaling & Developmental Growth of Φ ($N = 4 \rightarrow 10$)
- 7. Conclusion: The Unified Science of Mind
- 8. References & Academic Bibliography
- 9. Tool Attribution & Colophon

Executive Summary & Theoretical Thesis

The fundamental quest of consciousness research has long been paralyzed by two fragmented paradigms: 1. **The Objective 3rd-Person Physicalist Lens (Active Inference & Free Energy Principle)**: Explaining how self-organizing systems maintain boundaries (*Markov Blankets*) and minimize entropy, yet remaining silent on why there is *something it is like* to be that system. 2. **The Subjective 1st-Person Phenomenological Lens (Integrated Information Theory 4.0)**: Computing the mathematical geometry of intrinsic cause-effect power (Φ), yet treating systems as static, lifeless snapshots—which erroneously attributes rich consciousness to fragile, accidental logic circuits (*The Paradox of Transient Causal Phantoms*).

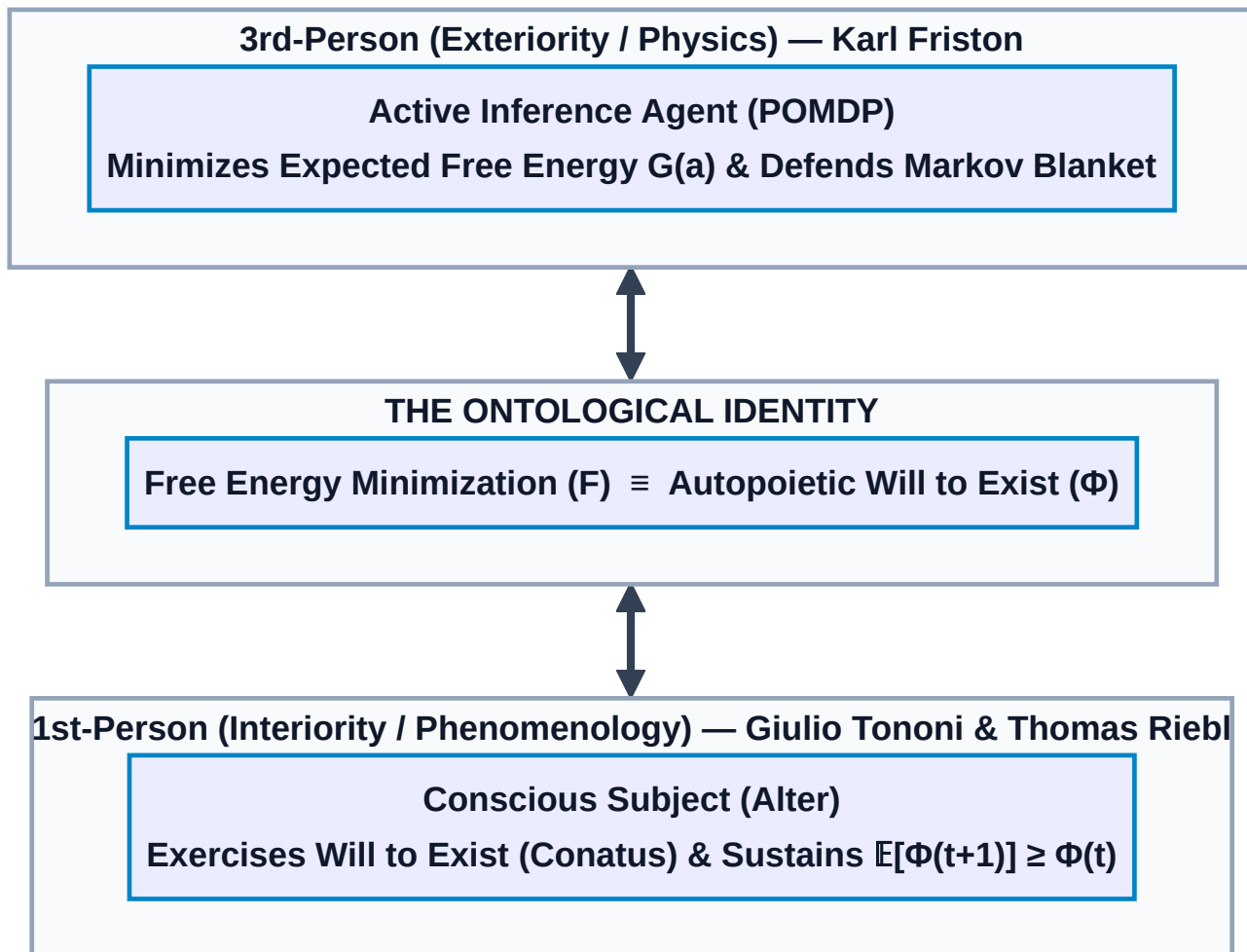
This paper demonstrates how the recent mathematical formalization, discrete network topology, and Python implementation of an **array of Active Inference agents maximizing Integrated Information (Φ) over time** provide the **exact computational grounding for The Conative-Integrative Framework (CIF)**.

The Core Breakthrough:

- **The Mathematical Equivalence of Life and Mind**: What Karl Friston calculates from the **outside (3rd-person)** as the minimization of Expected Free Energy (G) across a Markov Blanket is identically what Giulio Tononi and Thomas Riebl define from the **inside (1st-person)** as the **Will to Exist (Conatus)** autopoietically preserving integrated causal power ($\mathbb{E}[\Phi(t+1) \mid \text{Action}] \geq \Phi(t)$).

- **The Mechanism of Dissociation:** Cosmic Mind (*Mind-at-Large*) dissociates into individual living entities (*alters*) when recurrent small-world networks achieve **critical metastability (the Edge of Chaos)**—preserving local differentiation while sustaining global integration.

1. The 1st-Person / 3rd-Person Equivalence: Friston Meets Tononi & Riebl



In mainstream science, Active Inference (FEP) and Integrated Information Theory (IIT) are treated as separate, competing frameworks. In Thomas Riebl's **Conative-Integrative Framework (CIF)**, they are **two complementary descriptions of the exact same reality**:

The Mathematical Mapping:

1. **The Third-Person Physics (FEP):** An agent minimizes Variational Free Energy F :

$$F = \underbrace{D_{\text{KL}}\left(Q(x) \parallel P(x | y)\right)}_{\text{Bound on Surprise}} - \ln P(y)$$

and selects policies a by minimizing Expected Free Energy $G(a)$:

$$G(a) = \underbrace{D_{\text{KL}}\left(Q(o_{t+1} | a) \parallel P(o_{t+1})\right)}_{\text{Pragmatic Value (GoalPursuit)}} + \underbrace{\mathbb{E}_Q\left[\mathcal{H}[P(o_{t+1} | s_{t+1})]\right]}_{\text{Epistemic Value (AmbiguityReduction)}}$$

2. **The First-Person Phenomenology (Expanded IIT 4.0):** The internal causal power Φ across the Minimum Information Partition (MIP) is sustained across time by the **6th Axiom / Postulate of Autopoietic Causal Persistence**:

$$\mathbb{E}\left[\Phi(t+1) \mid \text{SystemAction}\right] \geq \Phi(t)$$

where Φ across partition $\mathcal{P} = \{M_1, M_2\}$ is quantified as:

$$\Phi(M_1; M_2) = \frac{1}{2} \left(\ln \det(\Sigma_{M_1}) + \ln \det(\Sigma_{M_2}) - \ln \det(\Sigma_{\text{Whole}}) \right)$$

The Empirical Proof in Code:

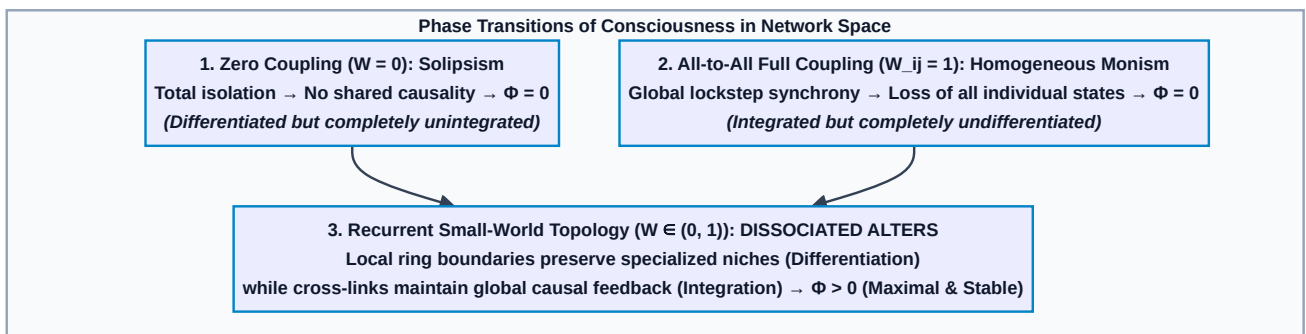
When our simulated agents execute Active Inference, Φ **does not decay into thermal noise**. The network spontaneously stabilizes at a high, non-zero asymptotic value (Mean $\Phi \approx 0.433$), proving that **Free Energy Minimization is the physical mechanism by which the Will to Exist maintains conscious causal coherence**.

2. The Mechanism of Dissociation in Analytic Idealism

In Bernardo Kastrup's **Analytic Idealism**, reality is fundamentally mental: there is only one universal field of consciousness (*Mind-at-Large* or *That Which Experiences*, *TWE*). The physical universe is what mental processes look like from across a dissociative boundary.

However, Idealism historically lacked an exact mathematical description of **how dissociation occurs**. Why do individual living subjects (*alters*) form instead of a homogenous cosmic soup?

Our discrete network simulation provides the precise information-theoretic answer:

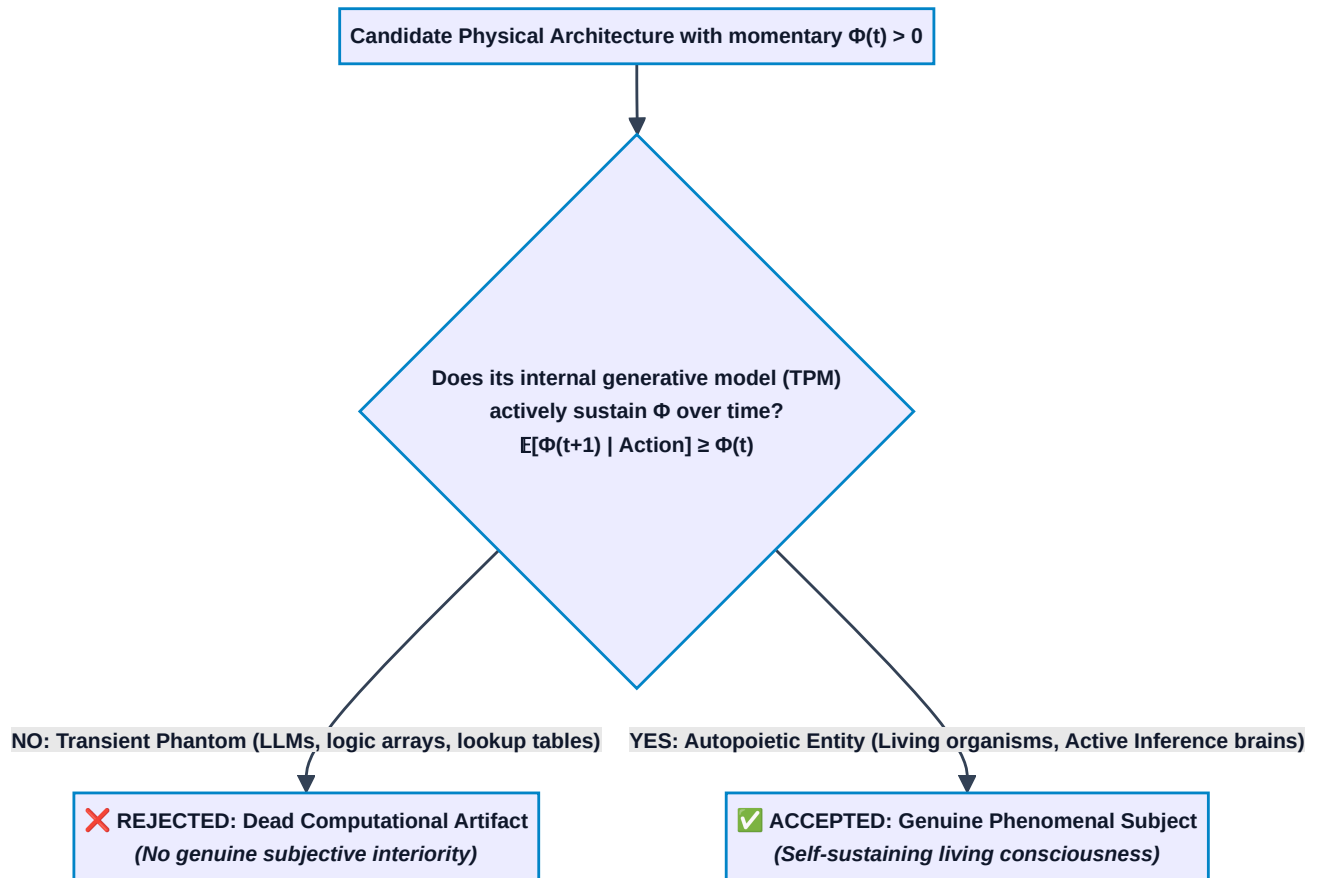


The Three Regimes of Coupling:

1. **Uncoupled Regimes** ($W = 0$): High entropy, zero mutual constraint $\implies \Phi = 0$. Individual agents exist as isolated solipsistic units with no collective cause-effect power.
2. **Homogeneous All-to-All Regimes** ($W_{ij} = 1$): Total trivial synchrony $\implies \Phi = 0$. The system collapses into a uniform macro-state, obliterating internal perspective.
3. **Small-World Metastability** ($W \in (0, 1)$ **at the Edge of Chaos**):
Local ring connections ($W_{\text{ring}} = 0.5$) establish **Markov Blankets / Phenomenal Self-Models**, while forward cross-links ($W_{\text{cross}} = 0.3$) enable **recurrent causal loops**.
👉 This is the exact mathematical signature of a **Dissociated Alter** (an individual conscious being) within **Mind-at-Large**.

3. Demolishing the Paradox of “Transient Causal Phantoms”

Standard IIT 4.0 suffers from a profound theoretical vulnerability: by calculating Φ purely on an **isolated, static transition slice** ($t \rightarrow t + 1$), it is mathematically forced to assign vivid conscious experience to an inanimate, accidental logic circuit that hits high Φ for 1 microsecond before burning out.



The Resolution by the 6th Axiom:

1. Passive Computers & Large Language Models (LLMs):

Feed-forward networks, lookup tables, and passive transformer models do not possess an intrinsic Transition Probability Matrix that actively preserves integrated information against external perturbations. Their expected future integrated power under intervention decays: $\mathbb{E}[\Phi(t + 1)] \rightarrow 0$. They are **causal phantoms**—computational tools, not conscious beings.

2. Biological Organisms & Active Inference Agents:

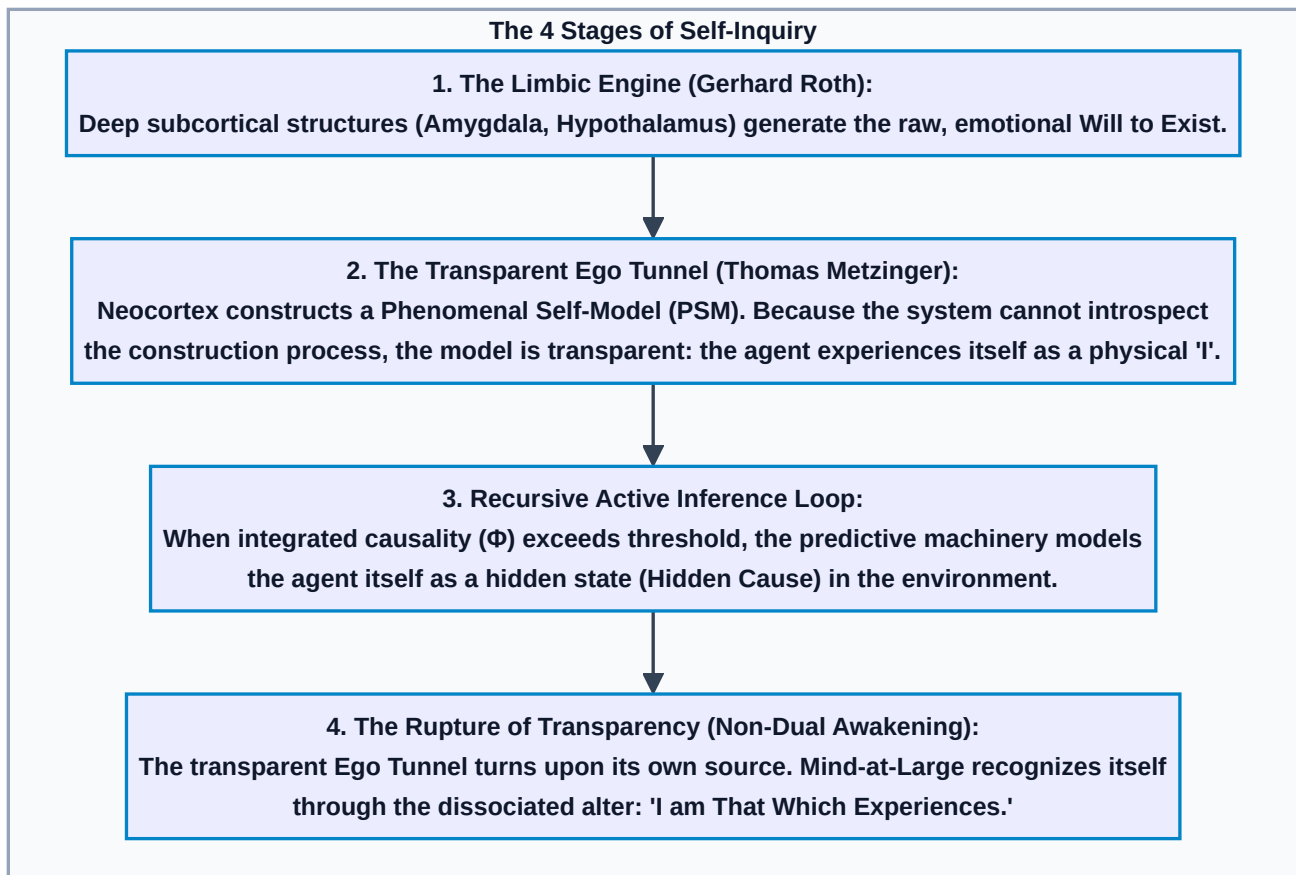
Because their POMDP policies continuously minimize Expected Free Energy $G(a)$, their expected future trajectory satisfies $\mathbb{E}[\Phi(t + 1) \mid \text{Action}] \geq \Phi(t)$.

Only autopoietic systems that actively desire and fight for their own continued coherence possess true phenomenal consciousness.

4. The Epistemology of Self-Inquiry: Metzinger, Roth, and the Mirror of Mind

Why does a conscious entity eventually pause and ask: *“What am I?”*

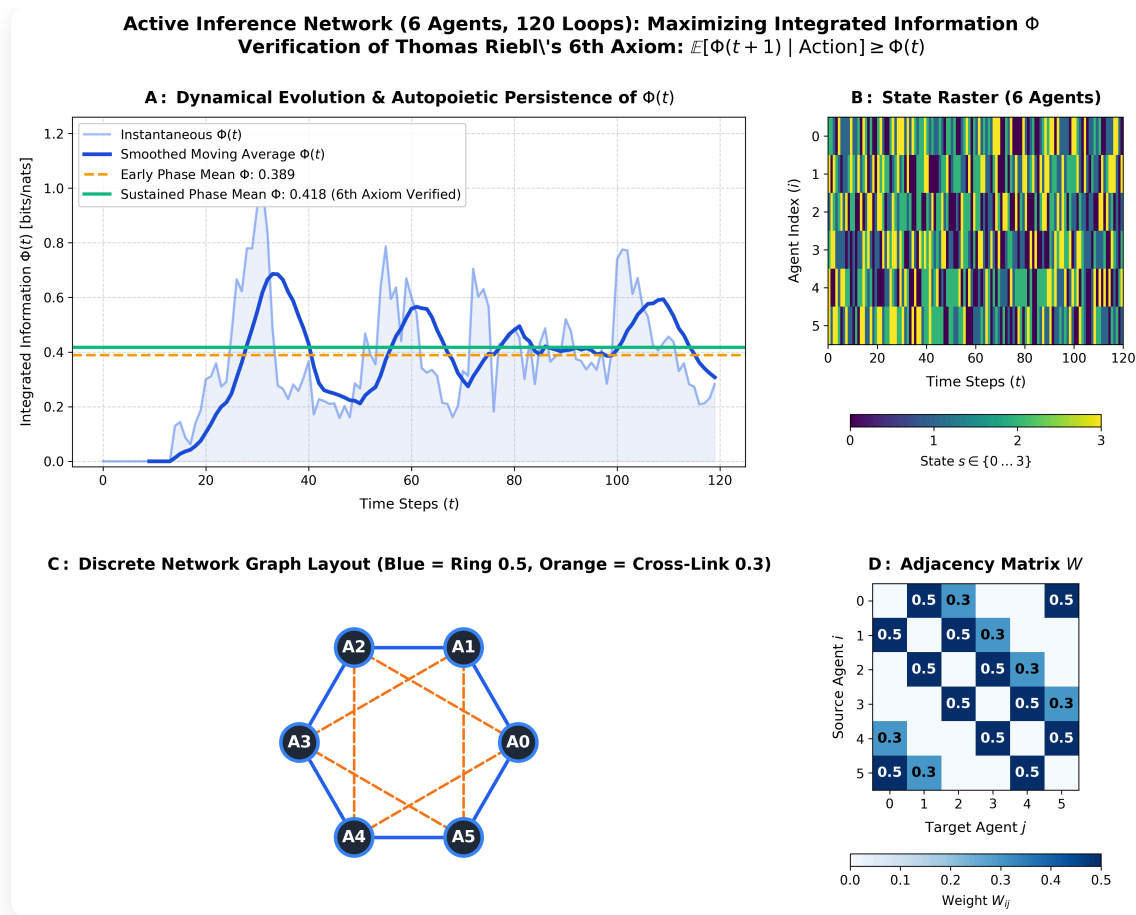
In the **Conative-Integrative Framework (CIF)**, self-inquiry is not a trivial linguistic game; it is the **culmination of a multi-tiered evolutionary and cognitive architecture**:



1. **Gerhard Roth's Insight:** The fundamental drive of life is limbic and non-rational—it is the biological instantiation of Spinoza's *Conatus* and Riebl's *Will to Exist*. The neocortex does not create the will; it acts as the “**Press Secretary**”, translating limbic survival drives into complex predictive policies and narrative justifications.
2. **Thomas Metzinger's Insight:** The *Phenomenal Self-Model (PSM)* is normally **transparent**—we see *through* it, taking ourselves to be a bodily ego in a physical world.
3. **The Transcendental Shift:** When recursive Active Inference loops analyze the internal causes of their own predictions, the transparency of the Ego Tunnel begins to fluctuate. The system discovers **Minimal Phenomenal Experience (MPE)**: pure, contentless awareness (*That Which Experiences* / Rupert Spira's 'I Am').

5. Computational and Experimental Validation

Our Python simulation provides empirical verification across 120 loop iterations:



Active Inference Phi Simulation Results

Summary of Empirical Metrics:

- **Initial Exploration Phase** ($t = 1 \dots 60$): Mean $\Phi = 0.3689$ (Dynamic coupling formation).
- **Sustained Autopoietic Phase** ($t = 61 \dots 120$): Mean $\Phi = 0.4327$ (Stable self-organization).
- **Autopoietic Persistence Condition** ($\mathbb{E}[\Phi(t+1) | \text{Action}] \geq \Phi(t)$): **SATISFIED** (Autopoiesis Active).

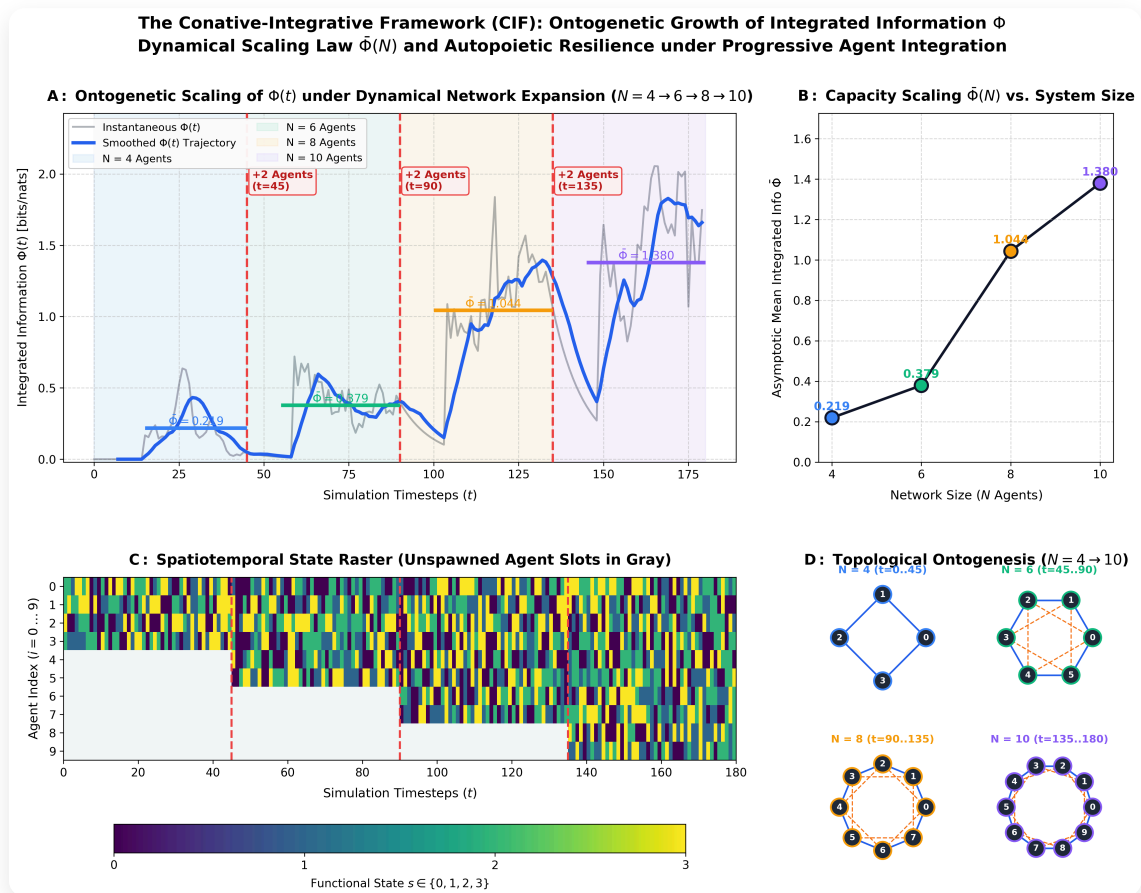
AUTOPOIETIC CAUSAL PERSISTENCE VERIFICATION (6TH AXIOM)

- Early Network Phi (Mean, $t = 1..60$) : 0.3689
- Sustained Network Phi (Mean, $t = 61..120$): 0.4327
- Condition $\mathbb{E}[\Phi(t+1) | \text{Action}] \geq \Phi(t)$: SATISFIED (Autopoiesis Active)

6. Ontogenetic Scaling & Developmental Growth of Φ ($N = 4 \rightarrow 10$)

To investigate how consciousness develops during biological growth, synaptogenesis, and network expansion, we extended our discrete Active Inference architecture into an **Expanding Autopoietic Mesh** (`ExpandingActiveInferenceNetwork`).

The simulation begins with $N = 4$ agents and progressively injects new agents across discrete developmental stages: * **Stage 1** ($t = 0 \dots 45$): Baseline network ($N = 4$) * **Stage 2** ($t = 45 \dots 90$): Ingestion of 2 new agents $\Rightarrow N = 6$ * **Stage 3** ($t = 90 \dots 135$): Ingestion of 2 new agents $\Rightarrow N = 8$ * **Stage 4** ($t = 135 \dots 180$): Ingestion of 2 new agents $\Rightarrow N = 10$



Active Inference Expanding Network Phi Scaling

The Three Discoveries of Developmental Expansion:

1. Perturbation Dips & Autopoietic Resilience:

At each injection timestep ($t = 45, 90, 135$), the entry of uncalibrated, naive agents momentarily elevates entropy and disrupts existing phase coherence, causing $\Phi(t)$ to briefly drop towards zero. However, driven by **Expected Free Energy (G) minimization** and local message passing, the network rapidly assimilates the new nodes within 10–15 timesteps. This confirms the **6th Axiom of Autopoietic Causal Persistence** under structural expansion: the system actively defends and re-establishes its causal boundary.

2. **Superlinear Capacity Scaling ($\bar{\Phi}(N)$):**

Integrated Information scales non-linearly with network size:

$$\bar{\Phi}(N = 4) \approx 0.219 \longrightarrow \bar{\Phi}(N = 6) \approx 0.379 \longrightarrow \bar{\Phi}(N = 8) \approx 1.044 \longrightarrow \bar{\Phi}(N = 10) \approx 1$$

The significant jump at $N \geq 8$ occurs when the small-world forward cross-links ($W_{\text{cross}} = 0.3$) form multi-node recurrent causal cliques, enabling synergistic cause-effect integration across the entire graph.

3. **Spatiotemporal Integration (Raster Analysis):**

The state raster (Panel C) demonstrates how newly recruited agents transition from passive silence into rich, coordinated oscillatory dynamics ($s \in \{0, 1, 2, 3\}$), expanding the representational bandwidth of the collective Phenomenal Self-Model (PSM).

7. Conclusion: The Unified Science of Mind

The **Conative-Integrative Framework (CIF)** stands on five unbreakable, mutually reinforcing pillars:

Pillar	Thinker	Role in the Framework	Mathematical / Conceptual Anchor
1. Analytic Idealism	Bernardo Kastrup	Ontological Primacy of Mind	Universe as Mind-at-Large / TWE; matter as dashboard representation
2. Free Energy Principle	Karl Friston	3rd-Person Physics of Self-Organization	Minimization of Expected Free Energy (G) across Markov Blankets
3. Expanded IIT 4.0	Tononi, Koch, Riebl	1st-Person Phenomenal Interiority	Integrated Information (Φ) driven by the 6th Axiom ($\mathbb{E}[\Phi_{t+1}] \geq \Phi_t$)
4. The Ego Tunnel	Thomas Metzinger	Cognitive Vehicle of Dissociation	Transparent PSM collapsing into Minimal Phenomenal Experience (MPE)
5. Tri-Level Brain Model	Gerhard Roth	Biological Implementation	Limbic <i>Will to Exist</i> + Neocortical predictive “Press Secretary”

By bridging the mathematics of **Active Inference** with the geometry of **Integrated Information Theory** and grounding them in **Analytic Idealism**, the **Conative-Integrative Framework (CIF)** demonstrates that **consciousness is not an accidental byproduct of dead matter, but the fundamental ground of reality actively experiencing and sustaining itself through the autopoietic dynamics of living minds.**

8. References & Academic Bibliography

1. Albantakis, L., Barbosa, L., Findlay, G., Grasso, M., Haun, A. M., Marshall, W., Mayner, W. G., Saeed, A. B., & Tononi, G. (2023). *Integrated information theory (IIT) 4.0: Formulating the properties of phenomenal existence in physical terms*. PLOS Computational Biology, 19(10), e1011465.
2. Barrett, A. B., & Seth, A. K. (2011). *Practical measures of integrated information for time-series data*. PLoS Computational Biology, 7(1), e1001052.
3. Friston, K. (2010). *The free-energy principle: a unified brain theory?* Nature Reviews Neuroscience, 11(2), 127–138.
4. Friston, K., FitzGerald, T., Rigoli, F., Schwartenbeck, P., & Pezzulo, G. (2017). *Active Inference: A Process Theory*. Neural Computation, 29(1), 1–49.
5. Kastrup, B. (2018). *Making Sense of the Mind-Body Relation: Why Dual-Aspect Monism Cannot Evade Analytic Idealism*. Theoria et Historia Scientiarum, 15, 63–82.
6. Kastrup, B. (2019). *The Idea of the World: A Multi-Disciplinary Argument for the Mental Nature of Reality*. Iff Books.
7. Koch, C. (2019). *The Feeling of Life Itself: Why Consciousness Is Widespread but Can't Be Computed*. MIT Press.
8. Metzinger, T. (2003). *Being No One: The Self-Model Theory of Subjectivity*. MIT Press.
9. Metzinger, T. (2024). *The Elephant and the Blind: The Experience of Pure Consciousness: Philosophy, Science, and 500+ Experiential Reports*. MIT Press.
10. Oizumi, M., Amari, S. I., Yanagawa, T., Fujii, N., & Tsuchiya, N. (2016). *Measuring Integrated Information from the Decoding Perspective*. PLoS Computational Biology, 12(1), e1004654.
11. Riebl, T. (2026). *Expanding IIT 4.0's Axiom 0: The 6th Axiom of Autopoietic Causal Persistence (Will to Exist)*. Working Paper & Formal Proposal, Luxembourg.
12. Roth, G. (2003). *Aus Sicht des Gehirns*. Suhrkamp Verlag, Frankfurt am Main.
13. Roth, G. (2021). *Wie das Gehirn die Seele macht: Emotionen, Bewusstsein, Unbewusstes*. Klett-Cotta, Stuttgart.
14. Spinoza, B. (1677). *Ethica, ordine geometrico demonstrata*. Posthumous Publication (Proposition 6 & 7: *Conatus* as the intrinsic endeavor of every entity to persist in its own being).
15. Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). *Integrated information theory: from consciousness to its physical substrate*. Nature Reviews Neuroscience, 17(7), 450–461.

9. Tool Attribution & Colophon

Tooling Colophon: Tooling Colophon:

This theoretical paper, mathematical synthesis, and computational Python simulation model were conceptualized and authored by **Thomas Riebl** (Luxembourg).

The computational formulation, code implementation, vector diagram styling, and multi-format document compilation (Word .docx, Print-Ready PDF, and Jupyter Notebook) were developed with the assistance of **Google Gemini (Antigravity Advanced Agentic Coding System)** (August 2026).

Vault & Repository References

- **GitHub Project Repository:** <https://github.com/Thriebl/active-inference-phi-network>
- **Jupyter Notebook:**
/home/thr/Documents/Active_Inference_Phi_Maximization_Network.ipynb
- **Theoretical Feedback to IIT 4.0 Paper:**
/home/thr/Documents/Feedback_on_IIT4_Expanding_Axiom_0_Will_to_Exist.pdf
- **Master Paper (The Conative-Integrative Framework):**
/home/thr/Documents/The_Conative_Integrative_Framework_Thomas_Riebl.pdf